SRIDEVI NIMMALA

+91-9014561612 — sridevinimmala13@gmail.com — Tadepalligudem, Andhra Pradesh
linkedin.com/in/sridevi-nimmala

Summary

ECE undergraduate passionate about Embedded Systems and RTL Design. Specifically interested in Embedded Hardware and Firmware development with hands-on experience in real-time embedded applications.

Education

Vishnu Institute of Technology, Bhimavaram — 2024 – Present
B.Tech in Electronics and Communication Engineering — CGPA: 8.8/10*

Sasi Institute of Technology, Tadepalligudem — 2021 – 2024
Diploma in Electronics and Communication Engineering — CGPA: 8.7/10

ZP High School, L. Agraharam — 2021
SSC Board — GPA: 9.2/10

Technical Skills

- **Programming Languages:** C, Python, Embedded C, ASP.NET.
- **Embedded Systems:** STM32, Raspberry(SMD RED), Arduino, Sensor Interfacing, Control system, UART, SPI, I2C Communication Protocols
- **Communication:** Optical Fiber Communication, Fiber Splicing Techniques, IP Phone Configuration
- **VLSI:** RTL Design, CMOS Layout Basics, Functional Verification, Testbench Development
- **Tools:** Wokwi, VS Code, Git, GitHub, MATLAB, Synopsis, Cadence Virtuoso, Xilinx Vivado, STM32CubeIDE, Arduino IDE

Projects

DC motor controlling using PID tuning (2026)

- Developing a real-time control system for tuning a DC motor using SMD RED raspberry pi and encoder.
- Working on sensor interfacing and implementing control algorithms for motor rotation and analyzing system response by doing python scripting.

Rotary Inverted Pendulum using STM32 (2026)

- Developing a real-time control system for stabilizing a rotary inverted pendulum using STM32 microcontroller.
- Working on sensor interfacing and implementing control algorithms for pendulum stabilization and analyzing system response in real time.

UART Communication Protocol Design using Verilog HDL (2025)

- Designed and implemented UART Transmitter and Receiver modules using Verilog HDL.
- Created testbench for functional verification and validated outputs through simulation.

Aqua Culture Monitoring System (2024)

- Designed an IoT-based monitoring system for aquaculture environments to measure pH, temperature, and oxygen levels with automated alerts.

IoT-Based Smart Agriculture System (2024)

- Implemented an automated irrigation system using soil moisture sensors and microcontroller-based control.

Internships & Workshops

Summer Internship – NIT (2026 – Present)

- Working on Embedded Systems and real-time hardware interfacing using STM32 microcontrollers.

VLSI Design Course – Internshala (2024)

- Learned RTL design concepts, Verilog HDL programming, testbench development, and functional verification.

Analog VLSI Design Workshop – Epic Layouts (2025)

- Designed CMOS layouts and performed DRC validation using Cadence Virtuoso.

VLSI Intern – NIT Tadepalligudem (2024)

- Studied switching systems and IP phone configuration with practical telecom exposure.

BSNL Telecommunications Internship (2024)

- Worked on optical fiber communication systems, communication protocols, and fiber splicing techniques.

Achievements

- 2nd Prize – Inter-College Hackathon (2024)
- Internshala VLSI – Elite Grade